Subpixel Spectral Variability Network for Hyperspectral Image Classification

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***Abstract*— Deep learning-based frameworks have shown great potential in the field of hyperspectral image (HSI) classification owing to their superior modeling capabilities. However, the existence of mixed pixels and spectral heterogeneity limits the discriminant performance of the classifier, which makes it impossible to distinguish the mixed spectra effectively in actual scenarios. To address this gap, we propose a subpixel spectral variability network (S^2VNet) for HSI classification, which incorporates complete subpixel information and class features modeled by spectral variability and nonlinear mixture characteristics to enhance classification performance. S^2VNet is capable of extracting endmembers and abundances based on the nonlinear autoencoder (AE) framework and estimating variability parameters by simultaneously considering scaling factors and perturbation terms to ensure accurate endmember construction. The enhanced subpixel fusion module is further designed to automatically integrate three aspects of abundances, spectral cosine correlation information, and pixel-level class features to provide a robust joint representation for the classifier. Extensive experiments on four public HSI datasets demonstrate the superiority and generalization of the proposed method when benchmarked with state-of-the-art methods. The code will be available at https://github.com/hanzhu97702/S2VNet.**

***Index Terms*— Image classification, nonlinear autoencoder (AE) network, remote sensing, spectral variability, subpixel information.**

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I. Introduction

HYPERSPECTRAL images (HSIs) provide a distinctive advantage over other remote sensing data sources and can precisely depict the spectral characteristics of objects [1], [2]. Owing to its fine spectral resolution, each pixel in the HSI can be regarded as a continuous spectral curve to provide abundant spectral information and identify different land-cover objects [3], [4]. HSI classification task aims at assigning a predefined semantic label to each pixel, so that the high-precision classification maps can be generated to reflect valuable distribution information on Earth's surface and has extensive applications in various fields, such as mineral exploration [5], [6], agriculture [7], [8], and environmental monitoring [9].

During the past decades, a series of improvements have been proposed in the field of HSI classification. Conventional HSI classification methods mainly focus on exploring machine learning techniques, such as the support vector machine (SVM) [10], random forest [11], and Bayesian model [12]. However, the extracted features by these classifiers fail to consider sufficient spatial information in the HSI. To effectively model spatial correlations in hyperspectral data, some advanced feature extraction approaches have been explored to improve the classification accuracy, including Gabor filtering [13], [14], morphological profiles [15], [16], and kernel-based analysis [17], [18], but they tend to require manual extraction of better features, which is not applicable in actual scenarios. Benefiting from strong modeling and feature learning capabilities, deep learning-based frameworks have shown great potential in the HSI classification task [19], [20]. Most representative network frameworks are based on convolutional neural networks (CNNs) and Transformers. In particular, CNNs aim at employing 2-D or 3-D convolution kernels to extract spectral–spatial features from the HSI [21], whereas the fixed receptive field enables CNNs to only retrieve local contextual information. Besides, several works try to design a hybrid network composed of 2-D and 3-D convolution operations to capture the discriminative spectral–spatial features [22], [23]. Different from the above methods, Transformers can effectively learn global information and long-distance dependencies by modeling the relationships between different pixels in the image and has shown impressive performance in image processing. To fully exploit spectral–spatial information of the HSI, a series of Transformer-based methods have shown pleasing competitiveness in classification performance [24],

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[25], [26]. Nevertheless, the Transformer-based approaches are generally insensitive to local information and the quadratic complexity of self-attention in Transformers also brings computational complexity problems [27], [28]. Therefore, many researchers tend to solve the imperfection of feature exploration by the hybrid architecture of CNNs and Transformers and greatly improve the classification performance in complex scenarios [29], [30].

Limited by spatial resolution and environmental influence in hyperspectral imaging, the existence of mixed pixels and spectral heterogeneity still hinder the discriminant performance of the classifier in existing pixel-level deep learning frameworks. The main reason is that the subpixel information in the HSI is usually ignored in the network design process. As a result, the trained classifier cannot distinguish the mixed spectra effectively in actual scenarios. To model subpixel information, Han et al. [31] first proposed a dual-branch subpixel-guided network (DSNet) for HSI classification and introduced an unsupervised autoencoder (AE) unmixing network to extract diagnostic abundances as prior subpixel features. Although dozens or even hundreds of spectral bands can provide rich spectral information for AE training, the extracted subpixel features do not contain the spectral variation characteristics, so that the classifier cannot comprehensively discriminate whether it is a pixel-level or subpixel-level feature category. In fact, spectral heterogeneity modeling is more complex, and learning discriminative spectral features is crucial for the HSI classification task [32]. The extracted pure spectra (i.e., endmembers) usually fluctuate with the acquisition conditions of each image, and spectral signatures of the same material at different spatial locations also display certain spectral variation patterns influenced by factors of illumination and terrain [33], [34]. There is neither a clear solution on how to model spectral heterogeneity from the HSI nor is it known what consequences there are of using extracted spectral variation characteristics for classifying HSI. Additionally, the high dimensionality of 3-D HSI data inevitably increases the spectral heterogeneity modeling difficulty, resulting in inefficient feature extraction and huge computational costs [35], [36]. Therefore, developing effective spectral heterogeneity modeling and subpixel information extraction techniques is crucial for improving the classification performance from the perspective of extending the generalization of spectral modeling.

Hyperspectral unmixing (HU) techniques offer a fine subpixel interpretation to deal with complex conditions in real scenarios. Recently, the data-driven AE unmixing model has become a landmark achievement in the HU field due to its unique self-supervised encoder–decoder architecture and excellent feature learning ability [37], [38], [39], [40]. Certain physical endmember and abundance network constraints enable these AE-based unmixing methods to intelligently obtain interpretable subpixel information from the HSI, which can further provide prior knowledge to serve downstream HSI tasks. However, existing AE-based unmixing methods also suffer from spectral variability, since most studies only consider the weights of the decoder as endmembers. To effectively construct spectral variation characteristics, there are two strategies to cope with spectral heterogeneity: extended endmember and variational inference. The extended endmember strategy usually adopts the constructed endmember library and collaborative sparsity representation to optimize coefficients of spectral heterogeneity [41]. Nevertheless, due to the manual setting of the number of spectral libraries, these priors cannot accurately characterize the intrinsic properties of HSIs and lead to poor generalization. The variational inference strategy tends to modify the decoder part by estimating the latent representation of endmembers and better model the variability [42]. However, this strategy may affect the acquisition of the endmember distribution because it focuses on scale variation of endmembers rather than random perturbations generated by the effects of the inherent variability of the material [43], [44]. Furthermore, the nonlinear interactions between different ground objects in the HSI affect the reconstruction quality of the deep learning-based frameworks during the training phase, which can result in restrictive modeling capacity [45], [46], [47], [48]. In this case, considering unified spectral modeling that incorporates more variation factors can further facilitate the generation of reliable subpixel information for the HSI classification task.

To this end, a novel subpixel spectral variability network (S^2VNet) is proposed to mitigate the aforementioned mixed pixel and spectral heterogeneity issues in the HSI classification task. S^2VNet is capable of effectively combining complete subpixel information and pixel-level class features to enhance HSI classification and generalization performance. Specifically, we design a general nonlinear AE unmixing framework to obtain subpixel information by incorporating spectral variability and nonlinear mixture characteristics simultaneously to reconstruct HSIs. The scaling factors and perturbation terms are independently modeled to learn the latent variation distribution so that the subpixel-level endmembers and abundances can be accurately estimated. The lightweight convolution and Transformer architecture is used for pixel-level class feature extraction at another flow of the proposed network. Furthermore, the enhanced subpixel fusion module is adopted to ensure better domain-invariant representation learning by applying the obtained abundance and endmember information synchronously to combine pixel-level class features for classifier discrimination. In brief, the major contributions of this article can be summarized as follows.

1) We present a subpixel S^2VNet for HSI classification to address the mixed pixel and spectral heterogeneity simultaneously, which provides rich multilevel information to enhance robust and generalization performance in actual scenarios.
2) The general nonlinear AE unmixing architecture is designed to extract accurate endmembers and abundances by effectively modeling multidimensional variation factors, including scaling, perturbation, and nonlinear interactions, to ensure the reconstruction quality.
3) The lightweight hybrid architecture based on the 2-D CNN and Transformer is adopted to obtain pixel-level class features independently of the HSI, providing stable class-wise representation for the subsequent fusion process.

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4) The enhanced subpixel fusion module is developed to achieve a comprehensive fusion of abundances, spectral cosine correlation information, and pixel-level class features, which further facilitates stable joint classification across subpixel-level and pixel-level information.

The remaining part of this article is organized as follows. Section II provides an introduction to related works in unmixing considering spectral variability and data-driven AE model. In Section III, we elaborate the implementation details of our S^{2}VNet. Section IV presents the extensive experiments and analyses on four public HSI classification datasets. Finally, conclusions are drawn in Section V.

II. Related Work

In this section, we first introduce the linear mixing model (LMM) and its extension for addressing spectral variability. The existing AE-based unmixing approaches are also presented in detail.

A. LMM With Spectral Variability

The LMM is a widely used spectral mixing model and assumes that each input pixel $\mathbf{y}_i$ in the HSI is constructed by the linear combination of endmembers and their fractional abundances, which can be denoted by

$$\begin{aligned} &\mathbf{Y} = \mathbf{MA} + \mathbf{N} \\ &\text{s.t. } \mathbf{1}_P^T\mathbf{A} = \mathbf{1}_N^T,\ \mathbf{A} \geq \mathbf{0},\ \ \mathbf{M} \geq \mathbf{0} \end{aligned} \tag{1}$$

where $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_N] \in \mathbb{R}^{L\times N}$ represents the input HSI with L spectral bands and N pixels. $\mathbf{M} = [\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_P] \in \mathbb{R}^{L\times P}$ is the endmember matrix with P ground object categories. $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_N] \in \mathbb{R}^{P\times N}$ is the abundance matrix and each abundance vector $\mathbf{a}_i$ should satisfy the abundance sum-to-one constraint (ASC) as well as the abundance nonnegativity constraint (ANC). $\mathbf{M}$ is also required to satisfy the endmember nonnegative constraint (ENC) to ensure physical realizability. $\mathbf{N} \in \mathbb{R}^{L\times N}$ denotes the additive noise matrix.

To effectively model spectral variability, the augmented LMM introduces scaling term and spectral variability library into LMM to provide an explicit physical consideration of spectral variabilities [49], which can be written as

$$\mathbf{Y} = \mathbf{MAS} + \mathbf{VB} + \mathbf{N} \tag{2}$$

where $\mathbf{S} \in \mathbb{R}^{N\times N}$ is a nonnegative diagonal matrix representing scalinng factors. $\mathbf{V} \in \mathbb{R}^{L\times D}$ denotes the constructed spectral variability library and D is the number of basic pure spectra. $\mathbf{B} \in \mathbb{R}^{D\times N}$ is the variability coefficient matrix corresponding to $\mathbf{V}$.

Another strategy based on variational inference is to learn the probability distribution of endmembers at each pixel position directly from the mixed data [42]. Since the true posterior distribution $p(\mathbf{M}|\mathbf{y})$ is intractable, a neural network $q_{\boldsymbol{\phi}_z}$ parameterized by $\boldsymbol{\phi}_z$ is designed to utilize the latent vector $\mathbf{z}$ conditioned on the pixel $\mathbf{y}$ as an approximate posterior distribution, which is expressed by a Gaussian as follows:

$$q_{\boldsymbol{\phi}_z}(\mathbf{z}|\mathbf{y}) = \mathcal{N}\left(\boldsymbol{\mu}_z(\mathbf{y}), \boldsymbol{\sigma}_z^2(\mathbf{y})\mathbf{I}\right) \tag{3}$$

where the latent vector $\mathbf{z}$ is reparameterized by $\mathbf{z} = \boldsymbol{\mu}_z(\mathbf{y}) + \boldsymbol{\sigma}_z(\mathbf{y}) \odot \boldsymbol{\varepsilon}$. $\mathbf{I}$ represents the identity matrix. Then, the scaling factor matrix $\mathbf{S}_v \in \mathbb{R}^{P\times P}$ of endmembers is estimated by implementing a nonlinear function f_θ as fully connected (FC) layers to assist the generation of endmembers. The process of endmember generation can be represented by the following equations:

$$\mathbf{S}_v = f_\theta(\mathbf{z}) \tag{4}$$

$$\mathbf{M}_v = \text{sigmoid}(\mathbf{MS}_v) \tag{5}$$

where $\mathbf{M}_v \in \mathbb{R}^{L\times P}$ represents the scaled endmembers after the imposition of scaling factors and its value spans from 0 to 1 via sigmoid activation. Based on the maximum likelihood estimation criterion and single Monte Carlo sampling, the training process of the variational inference strategy can be approximated by optimizing the reconstruction and Kullback–Leibler (KL) divergence loss as

$$\begin{aligned} \mathcal{L}\left(\mathbf{y}; \boldsymbol{\phi}_z, \boldsymbol{\theta}\right) &= \mathbb{E}_{q_{\boldsymbol{\phi}_z}(\mathbf{z}|\mathbf{y})}\left[\log p_\theta(\mathbf{y}|\mathbf{z})\right] - D_{\text{KL}}\left(q_{\boldsymbol{\phi}_z}(\mathbf{z}|\mathbf{y})||p(\mathbf{z})\right) \\ &\propto -||\mathbf{y} - \mathbf{M}_v\mathbf{A}||_F^2 - D_{\text{KL}}\left(q_{\boldsymbol{\phi}_z}(\mathbf{z}|\mathbf{y})||p(\mathbf{z})\right) \end{aligned} \tag{6}$$

B. AE-Based Unmixing Approaches

As a typical representative of the deep learning framework, AE can achieve unsupervised unmixing performance by imposing certain physical constraints on the encoder–decoder architecture. The encoder part in AE aims to learn the transformation from the input pixel to the hidden representation, namely abundances since the abundances share the same spatial structure as the HSI but with much lower dimension. The encoding process of AE can be formulated as

$$\hat{\mathbf{A}} = f_E(\mathbf{Y}) = \sigma\left(\mathbf{W}^{(e)}\mathbf{Y} + \mathbf{b}^{(e)}\right) \tag{7}$$

where $\hat{\mathbf{A}} \in \mathbb{R}^{P\times N}$ is the learned abundance results. $\sigma(\cdot)$ denotes the nonlinear activation function, such as the rectified linear unit (ReLU) and the leaky ReLU (LReLU). $\{\mathbf{W}^{(e)}, \mathbf{b}^{(e)}\}$ represents the set of the weight and bias matrix in the eth encoder part.

Based on the LMM assumption, the decoder part in AE mainly adopts one FC layer to reconstruct the input HSI from abundances, which can be written as

$$\hat{\mathbf{Y}} = f_D\left(\hat{\mathbf{A}}\right) = \mathbf{W}^{(d)}\hat{\mathbf{A}} \tag{8}$$

where $\hat{\mathbf{Y}} \in \mathbb{R}^{L\times N}$ is the reconstructed HSI. $\mathbf{W}^{(d)} \in \mathbb{R}^{L\times P}$ represents the weight matrix in the decoder part, corresponding to the estimated endmember results in $\hat{\mathbf{M}}$. Endmembers refer to the pure spectral characteristics of different materials present in the scene, which can preserve invariant spectral priors in the HSI.

III. Methodology

A. Overview of S^{2}VNet

As illustrated in Fig. 1, our S^{2}VNet consists of three parts: the general nonlinear AE unmixing network, the pixel-level classifier network, and the enhanced subpixel fusion module. The general nonlinear AE unmixing network aims at generating reliable subpixel information by effectively modeling

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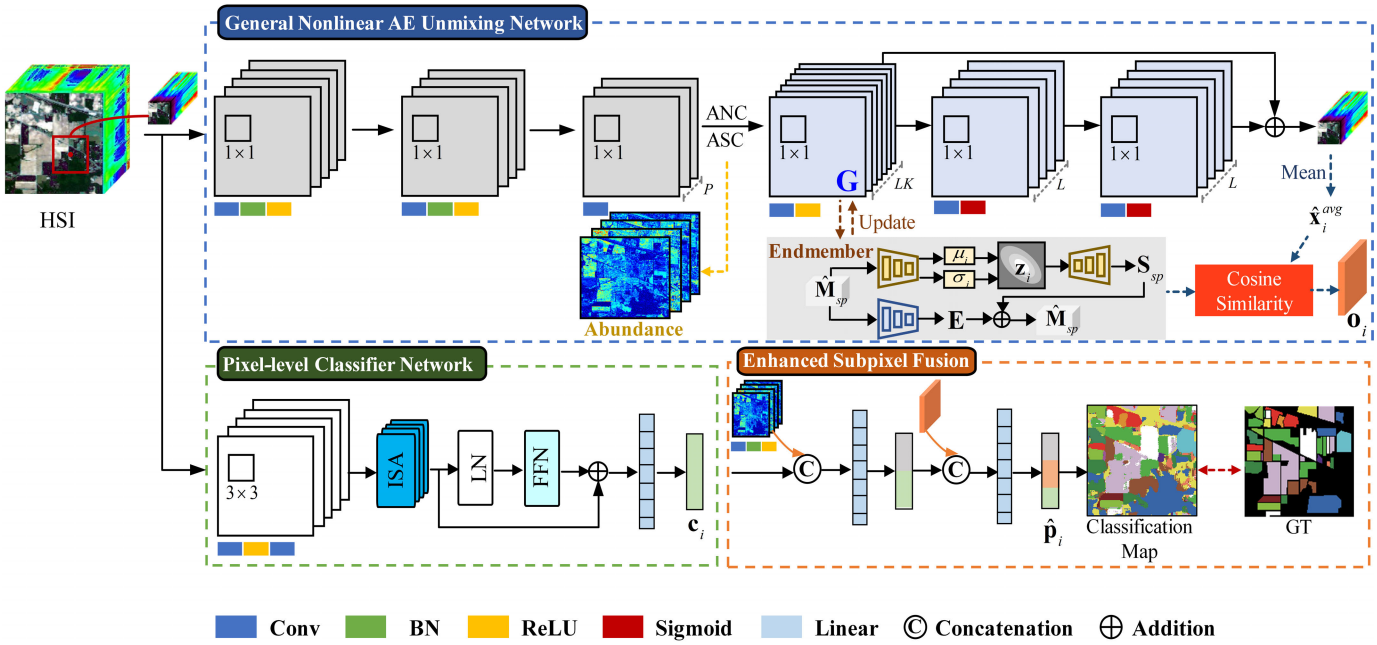


Fig. 1. Framework of the proposed S^{2}VNet, including general nonlinear AE unmixing network, pixel-level classifier network, and enhanced subpixel fusion module.

spectral variability and nonlinear mixture characteristics to reconstruct the HSI in an unsupervised manner. The pixel-level classifier network adopts the lightweight 2-D CNN and Transformer architecture to provide pixel-level class features. The enhanced subpixel fusion module is finally used to integrate the extracted subpixel information and pixel-level class features to realize domain-invariant representation learning.

B. General Nonlinear AE Unmixing Network

Owing to the existence of multiple scattering interactions between different materials, LMM cannot hold the spectral mixing mechanism in real scenarios, while the nonlinear mixing model (NLMM) can approximate the actual mixing situation by modeling the complicated interactions of ground objects with height differences, such as building and tree. The general form of NLMM can be expressed as the sum of a linear mixture and a nonlinear fluctuation parameterized by endmembers and corresponding abundances [50], [51], that is

$$\mathbf{Y} = \mathbf{MA} + \Phi(\mathbf{MA}) + \mathbf{N} \tag{9}$$

where Φ denotes the nonlinear mapping function applied to the linear transform $\mathbf{MA}$. Our previous work has demonstrated the feasibility of the data-driven NLMM-based AE framework in the HSI classification task [31]. Given the input patch $\{\mathbf{x}_i\}_{i=1}^{N_S} \in \mathbb{R}^{L\times H\times W}$ with the $H \times W$ spatial size and N_S training samples, the reconstruction process of the general mixing decoder in the NLMM-based AE framework can be formulated as

$$\hat{\mathbf{x}}_i = f_D(\mathbf{v}_i) = \sum_{k=1}^{K} \mathbf{G}_k \mathbf{v}_i + \mathcal{T}_{LK\to L}(\mathbf{G}\mathbf{v}_i) \tag{10}$$

where $\hat{\mathbf{x}}_i \in \mathbb{R}^{L\times H\times W}$ is the reconstructed patch obtained from NLMM-based AE framework. $\mathbf{G} \in \mathbb{R}^{LK\times P}$ is the designed weight matrix to unify the relationship between the number of decoder layers K and different mixing modeling, which can further obtain endmember results by $\hat{\mathbf{M}} = \sum_{k=1}^{K} \mathbf{G}_k$. $\mathcal{T}_{LK\to L}(\cdot)$ stands for the nonlinear mapping function that transforms features from the LK dimension space to the L dimension space to describe nonlinear interactions. $\mathbf{v}_i \in \mathbb{R}^{P\times H\times W}$ denotes the extracted abundances from the encoder part as (7). The ANC and ASC are satisfied by applying the absolute value rectification and summed normalization operations as follows:

$$\mathbf{v}_i = \frac{|\mathbf{h}_i|}{\sum_{i=1}^{P} |\mathbf{h}_i|} \tag{11}$$

where $\mathbf{h}_i \in \mathbb{R}^{P\times H\times W}$ represents the encoded hierarchical representation of the input HSI in the encoder part.

To deal with spectral heterogeneity, the spectral variability characteristics in the HSI are modeled by introducing scaling factors and perturbation terms simultaneously to ensure accurate endmember construction defined as

$$\hat{\mathbf{M}}_{sp} = \text{sigmoid}\left(\sum_{k=1}^{K} \mathbf{G}_k \mathbf{S}_{sp} + \mathbf{E}\right) \tag{12}$$

where $\mathbf{S}_{sp} \in \mathbb{R}^{P\times P}$ is the scaling factor matrix and $\mathbf{E} \in \mathbb{R}^{L\times P}$ is the perturbation term to learn random perturbations generated by the inherent variability of different materials. More specifically, the scaling factor matrix is generated based on variational inference and the latent vector parameterized with the sample mean $\boldsymbol{\mu}_i$ and covariance $\boldsymbol{\sigma}_i$ based on Gaussian distribution, that is

$$p(\mathbf{z}_i|\hat{\mathbf{m}}_i) = \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\sigma}_i^2\mathbf{I}),\ \boldsymbol{\mu}_i = f_{\theta_1}(\hat{\mathbf{m}}_i),\ \boldsymbol{\sigma}_i = f_{\theta_2}(\hat{\mathbf{m}}_i) \tag{13}$$

where $f_{\theta_1}(\cdot)$ and $f_{\theta_2}(\cdot)$ are two FC layers. Then, the reparameterization trick [52] is adopted to conduct the sampling

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operations as follows:

$$\mathbf{z}_i = \boldsymbol{\mu}_i + \boldsymbol{\sigma}_i \odot \boldsymbol{\varepsilon},\ \boldsymbol{\varepsilon} = \mathcal{N}(0, \mathbf{I}) \tag{14}$$

where $\mathbf{z}_i \in \mathbb{R}^{N_L}$ is the learned latent vector and can further obtain $\mathbf{S}_{sp}$ as (4) by applying two FC layers to convert the dimension from N_L to P^2.

The generation process of $\mathbf{E}$ is designed as

$$\mathbf{E} = f_{\text{perturb}}\left(\hat{M}\right) = \sigma\left(\sum_{k=1}^{K} \mathbf{G}_k \mathbf{W}^{(z)} + \mathbf{b}^{(z)}\right) \tag{15}$$

where $f_{\text{perturb}}(\cdot)$ denotes the perturbation estimation network by one FC layer and ReLU activation function. We assume that the perturbation term has a certain addictive range by applying the ReLU function as the final output layer. $\mathbf{W}^{(z)} \in \mathbb{R}^{P\times P}$ and $\mathbf{b}^{(z)} \in \mathbb{R}^{L\times P}$ are the weight and bias matrix in the perturbation estimation network, respectively. In each iteration, the new endmember results $\hat{\mathbf{M}}_{sp}$ are reassigned to the general nonlinear AE unmixing network to achieve the synchronous training of spectral variability and nonlinear mixture characteristics. Note that the proposed unmixing network does not need to preassign an initial value of endmembers to the network, for example, vertex component analysis, since the decoder part covers rich spectral modeling.

C. Pixel-Level Classifier Network

The motivation of the pixel-level classifier network is to provide pixel-level class information and does not bring more computational burden. Here, inspired by decomposed self-attention [53], the lightweight hybrid architecture based on the 2-D CNN and Transformer is adopted as a pixel-level classifier network. First, the convolution tokenization (CT) is designed to encoder robust contextual features from the HSI, including two 3×3 convolution layers with the ReLU activation function, and the process can be described as

$$\mathbf{d}_i = \text{Reshape}(\text{CT}(\mathbf{x}_i)) \tag{16}$$

where $\mathbf{d}_i \in \mathbb{R}^{C\times H\times W}$ is the encoded patch token with C embedding dimension.

Owing to the quadratic complexity of self-attention in Transformers, the improved self-attention (ISA) in the decomposition form is formulated as follows:

$$\mathbf{A}_H = \text{Softmax}\left(\mathbf{Q}_H \mathbf{K}_H^T\right) \tag{17}$$

$$\mathbf{A}_W = \text{Softmax}\left(\mathbf{Q}_W \mathbf{K}_W^T\right) \tag{18}$$

$$\mathbf{head}_i = \text{ISA}(\mathbf{d}_i) = \mathbf{A}_H(\mathbf{A}_W \mathbf{V}_W)^T \tag{19}$$

where $\mathbf{A}_H \in \mathbb{R}^{H\times H}$ and $\mathbf{A}_W \in \mathbb{R}^{W\times W}$ are the independent horizontal and vertical adjacency matrix. $\mathbf{Q}_R = \mathbf{d}_R \mathbf{W}^Q$, $\mathbf{K}_R = \mathbf{d}_R \mathbf{W}^K$, and $\mathbf{V}_R = \mathbf{d}_R \mathbf{W}^V$ represent the region-level key, query, and value for each head along the specific spatial dimension, respectively. $\mathbf{d}_R \in \mathbb{R}^{R\times C}$ is reshaped from $\mathbf{d}_i$ to only compute spatial adjacency in the R dimension. R stands for H or W. $\mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V \in \mathbb{R}^{C\times(C/h)}$ are learnable projection weights in h heads. Then, the output of each head is concatenated to perform a multihead operation followed by layer normalization (LN) and feed-forward network (FFN), which can be represented as

$$\mathbf{F}_i = \text{Concat}(\mathbf{head}_1, \ldots, \mathbf{head}_h)\mathbf{W}^o \tag{20}$$

$$\bar{\mathbf{F}}_i = \text{FFN}(\text{LN}(\mathbf{F}_i)) + \mathbf{F}_i \tag{21}$$

where $\mathbf{F}_i$ and $\bar{\mathbf{F}}_i$ are the outputs of the Transformer and the FFN module. $\mathbf{W}^o \in \mathbb{R}^{C\times C}$ is adopted to compose all heads. The pixel-level class features $\mathbf{c}_i \in \mathbb{R}^P$ are obtained via a classifier head, including an average pooling operation and one FC layer

$$\mathbf{c}_i = f_{ch}\left(\bar{\mathbf{F}}_i\right) \tag{22}$$

where $f_{ch}(\cdot)$ denotes the transformation process in the classifier head.

D. Enhanced Subpixel Fusion Module

To effectively aggregate subpixel and pixel information, the enhanced subpixel fusion module is designed to achieve comprehensive fusion from three aspects of abundances, spectral cosine correlation information, and pixel-level class features to facilitate the supervised training of S^{2}VNet. Here, endmembers and abundances extracted by a general nonlinear AE unmixing network are simultaneously considered as subpixel prior knowledge for the classifier's discriminant performance learning. Specifically, abundances focus more on the description of the spatial distribution differences at the subpixel level, while endmembers tend to consider the class correlation of the reconstructed image at the pixel level. Since the obtained endmember results focus on the curve variations of various ground objects, the spectral cosine correlation information is adopted to reflect cosine similarities between the reconstructed pixel $\hat{\mathbf{x}}_i$ and endmembers $\hat{\mathbf{M}}_{sp}$, that is

$$\mathbf{o}_i = \frac{\hat{\mathbf{x}}_i^{\text{avg}} \hat{\mathbf{M}}_{sp}}{||\hat{\mathbf{x}}_i^{\text{avg}}||_2 ||\hat{\mathbf{M}}_{sp}||_2} \tag{23}$$

where $\mathbf{o}_i \in \mathbb{R}^P$ is the generated spectral cosine correlation information and each element ranges from -1 to 1. $\hat{\mathbf{x}}_i^{\text{avg}} \in \mathbb{R}^L$ represents the averaged reconstruction pixel obtained from general nonlinear unmixing network.

To enhance the functionality of each part representation, we adopt a two-stage fusion strategy to integrate diverse discriminant features to optimize the class decision process

$$\hat{\mathbf{p}}_i = f_{\text{out2}}\left(\text{Concat}\left(\sigma\left(f_{\text{out1}}\left(\text{Concat}\left(\mathbf{v}_i^{\text{conv}}, \mathbf{c}_i\right)\right)\right), \mathbf{o}_i\right)\right) \tag{24}$$

where $\hat{\mathbf{p}}_i \in \mathbb{R}^P$ represents the output class prediction by S^{2}VNet. $\mathbf{v}_i^{\text{conv}}$ is the flattened abundance vector after 3×3 convolution layer with a stride of 2. $f_{\text{out1}}(\cdot)$ and $f_{\text{out2}}(\cdot)$ are independent FC layer to transform the concatenated feature dimension into the dimension of the class. Note that, this two-stage fusion strategy can flexibly learn the unique class information in different HSI scenarios and obtain better class discrimination results.

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E. Objective Function

The overall objective function of S^{2}VNet includes the dual-branch training of a general nonlinear AE unmixing network and pixel-level classifier network. The reconstruction (RE) loss based on spectral angle distance (SAD) and KL divergence loss are utilized to train the general nonlinear AE unmixing network by imposing nonlinear mixture and spectral variability characteristics, given by

$$L_{\mathrm{RE}} = \frac{1}{N_S}\sum_{i=1}^{N_S}\arccos\left(\frac{\hat{\mathbf{x}}_i^T\mathbf{x}_i}{\|\hat{\mathbf{x}}_i\|_2\|\mathbf{x}_i\|_2}\right) \tag{25}$$

$$L_{\mathrm{KL}} = D_{\mathrm{KL}}\left(p\left(\mathbf{z}_i|\hat{\mathbf{m}}_i\right)||p(\mathbf{z})\right) = \sum_{i=1}^{N_L}-\frac{1}{2}\left[\log\boldsymbol{\sigma}_i - \boldsymbol{\mu}_i^2 - \boldsymbol{\sigma}_i^2 + 1\right] \tag{26}$$

where $\mathbf{x}_i$ and $\hat{\mathbf{x}}_i$ are the ith input patch and reconstructed patch in the HSI, respectively. N_L denotes the dimension of the latent vector. $p(\mathbf{z})$ is the prior distribution of the latent vector and is assumed to obey the Gaussian distribution.

Moreover, S^{2}VNet takes into account the continuity of ground object distribution in space, and the total variation (TV) smooth constraints are imposed on the abundance and spectral perturbation term, which can be written as

$$L_{\mathrm{TV_{abu}}} = \frac{1}{2HW}\sum_{i=1}^{H-1}\sum_{j=1}^{W-1}\left(\left(\mathbf{v}_{:,i,j+1}-\mathbf{v}_{:,i,j}\right)^2+\left(\mathbf{v}_{:,i+1,j}-\mathbf{v}_{:,i,j}\right)^2\right) \tag{27}$$

$$L_{\mathrm{TV_{per}}} = \frac{1}{P}\sum_{i=1}^{L-1}\|\mathbf{E}_{i,:}-\mathbf{E}_{i+1,:}\|_F^2. \tag{28}$$

The training of the pixel-level classifier network adopts the cross-entropy (CE) loss between the output class prediction and the true label. The standard CE loss is defined as

$$L_{\mathrm{CE}} = -\frac{1}{N_S}\sum_{i=1}^{N_S}\mathbf{p}_i\log\left(\hat{\mathbf{p}}_i\right) \tag{29}$$

where $\mathbf{p}_i$ and $\hat{\mathbf{p}}_i$ are the one-hot encoding of the true label in the ground truth (GT) and the corresponding class prediction of S^{2}VNet. Finally, the joint classification loss of S^{2}VNet can be summarized as

$$L = \underbrace{L_{\mathrm{CE}}}_{\text{classifier loss}} + \underbrace{L_{\mathrm{RE}}+\lambda_1 L_{\mathrm{KL}}+\lambda_2 L_{\mathrm{TV_{abu}}}+\lambda_3 L_{\mathrm{TV_{per}}}}_{\text{unmixing loss}} \tag{30}$$

where λ_1, λ_2, and λ_3 are the independent regularization parameters associated with KL divergence loss and TV loss to balance the significance of each part and achieve effective discriminative performance learning. Note that the input of the network is the image patch generated by taking the labeled training sample points as the center point to extract the spatial size of $H \times W$ and L bands. The entire network architecture does not require distributed training and it only needs to optimize (30) to jointly train these two branches.

IV. Experiments

In this section, the experimental settings are briefly presented and the corresponding quantitative analyses are carried out to demonstrate the effectiveness of the proposed method. Eight mainstream classification approaches are adopted to evaluate classification performance on four HSI baseline datasets, including the 2-D CNN [54], 3-D CNN [21], Vision Transformer (ViT) [55], morphological convolutional network (MorphConv) [56], spectral–spatial feature tokenization Transformer (SSFTT) [25], weighted feature fusion of the CNN and the graph attention network (WFCG) [57], DSNet [31], and the hybrid transformer network (HybridFormer) [29]. To provide intuitive measurement criteria, the overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa) are adopted to evaluate classification performance for different HSI classification methods.

A. Dataset Description and Experimental Setup

1) Indian Pines Dataset: The Indian Pines dataset was acquired by the airborne visible infrared imaging spectrometer (AVIRIS) [58] over a study region located in Northwest Indiana, USA. It consists of 145 × 145 pixels with a spatial resolution of 20 m and 220 spectral bands ranging from 400 to 2500 nm. After removing the water absorption and noise bands, only 200 bands remained with 16 classes, as detailed in Table I.

2) Berlin Dataset: The Berlin dataset was collected by the environmental mapping and analysis program (EnMAP)[1] over the city of Berlin, Germany, and the spatial resolution is 30 m. This dataset has a size of 1723 × 476 pixels with 244 spectral bands covering the wavelength range from 400 to 2500 nm. Eight classes of land cover are considered and the sample details are shown in Table II.

3) Augsburg Dataset: The Augsburg dataset was gathered by the HySpex sensor [59] over the city of Augsburg, Germany. It has a spatial resolution of 30 m with a size of 332 × 485 pixels and 180 available spectral bands ranging from 400 to 2500 nm. This dataset encompasses seven categories and specific information regarding the land-cover types is provided in Table III.

4) Cross-Scene Houston Dataset: The cross-scene Houston dataset was released in 2013 and 2018 by IEEE GRSS DFC to validate the generalization performance of different HSI classification approaches [60]. It is composed of 209 × 955 pixels and 48 spectral bands in the selected overlapping area of the Houston scene. Due to the large time span, this dataset can be used to evaluate the generalization performance of different classification methods, and the corresponding spatial resolution is uniformly resampled to 2.5 m. There are seven consistent classes involved in this dataset and Table IV lists the number of training and test samples.

The hardware environment adopted for experiments is NVIDIA GTX 1080Ti 11-GB GPU. For a fair comparison, all comparison approaches are executed based on the PyTorch framework and adopt the same hyperparameters in the original

[1] http://doi.org/10.5880/enmap.2016.002.

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TABLE I

CLASS-SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS ON THE INDIAN PINES DATASET

Class Name	Training Samples	Test Samples	Classic Backbone Networks			Hybrid Backbone Networks					
			2-D CNN [54]	3-D CNN [21]	ViT [55]	MorphConv [56]	SSFTT [25]	WFCG [57]	DSNet [31]	HybridFormer [29]	S^2VNet
Corn Notill	50	1384	77.24	65.25	66.55	71.53	85.62	80.92	83.96	85.62	**93.35**
Corn Mintill	50	784	84.57	59.57	88.52	84.57	90.43	90.69	94.90	**99.23**	96.56
Corn	50	184	92.39	94.57	88.59	97.83	97.28	**100.00**	99.46	**100.00**	99.46
Grass Pasture	50	447	94.63	89.71	97.09	93.74	90.60	94.41	**98.21**	97.76	96.20
Grass Trees	50	697	78.62	95.12	85.65	92.25	95.95	99.43	97.27	**99.57**	**99.57**
Hay Windrowed	50	439	92.26	98.63	98.86	98.86	98.63	**100.00**	99.54	**100.00**	**100.00**
Soybean Notill	50	918	79.63	79.74	93.03	73.31	85.51	82.79	95.10	94.01	**97.39**
Soybean Mintill	50	2418	67.74	67.70	73.95	56.95	75.27	87.68	90.07	84.82	**92.39**
Soybean Clean	50	564	74.29	72.87	75.18	81.91	78.90	86.70	94.68	95.04	**95.21**
Wheat	50	162	99.38	99.38	99.38	**100.00**	**100.00**	**100.00**	99.38	**100.00**	**100.00**
Woods	50	1244	92.44	79.98	92.04	87.70	94.21	**99.92**	95.90	96.54	97.03
Buildings Grass Trees Drives	50	330	83.94	67.58	89.39	93.33	92.73	98.18	97.58	99.39	**99.70**
Stone Steel Towers	50	45	**100.00**	**100.00**	**100.00**	**100.00**	97.78	**100.00**	**100.00**	**100.00**	**100.00**
Alfalfa	15	39	74.36	94.87	92.31	**100.00**	94.87	97.44	94.87	**100.00**	97.44
Grass Pasture Mowed	15	11	**100.00**	**100.00**	**100.00**	90.91	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
Oats	15	5	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
OA (%)			80.07	75.46	82.79	77.56	86.51	90.64	93.08	92.63	**95.72**
AA (%)			86.97	85.31	90.03	88.93	92.33	94.88	96.31	97.00	**97.77**
Kappa (%)			77.43	72.13	80.44	74.65	84.63	89.27	92.08	91.58	**95.10**

TABLE II

CLASS-SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS ON THE BERLIN DATASET

Class Name	Training Samples	Test Samples	Classic Backbone Networks			Hybrid Backbone Networks					
			2-D CNN [54]	3-D CNN [21]	ViT [55]	MorphConv [56]	SSFTT [25]	WFCG [57]	DSNet [31]	HybridFormer [29]	S^2VNet
Forest	443	54511	**77.16**	73.10	66.74	55.24	47.84	76.46	75.44	32.47	75.11
Residential Area	423	268219	66.33	55.27	57.86	72.47	73.58	68.80	74.69	81.11	**81.13**
Industrial Area	499	19067	21.37	43.92	**55.85**	26.25	30.59	42.06	54.91	42.78	49.37
Low Plants	376	58906	69.11	68.76	76.25	74.27	84.73	67.58	81.79	**90.08**	84.61
Soil	331	17095	68.15	84.05	80.04	84.88	92.20	**97.91**	74.19	64.79	72.49
Allotment	280	13025	58.52	32.48	64.17	61.21	49.21	**72.80**	63.71	54.76	49.46
Commercial Area	298	24526	47.47	30.50	29.90	49.90	39.25	**53.21**	30.56	25.64	32.80
Water	170	6502	65.70	58.27	**78.05**	65.47	63.63	71.16	67.44	74.12	66.03
OA (%)			64.94	57.77	60.97	67.60	68.23	68.83	72.09	70.54	**75.56**
AA (%)			59.23	55.80	63.61	61.21	60.13	**68.75**	65.34	58.22	63.88
Kappa (%)			50.06	42.84	47.28	53.38	53.73	56.48	59.27	55.10	**62.95**

TABLE III

CLASS-SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS ON THE AUGSBURG DATASET

Class Name	Training Samples	Test Samples	Classic Backbone Networks			Hybrid Backbone Networks					
			2-D CNN [54]	3-D CNN [21]	ViT [55]	MorphConv [56]	SSFTT [25]	WFCG [57]	DSNet [31]	HybridFormer [29]	S^2VNet
Forest	146	13361	85.76	93.02	82.44	93.21	95.52	91.65	93.19	96.45	**98.44**
Residential Area	264	30065	87.44	92.68	88.77	95.83	96.39	**98.96**	95.85	95.00	95.27
Industrial Area	21	3830	69.56	21.46	**79.27**	32.69	25.93	50.97	62.85	51.62	74.23
Low Plants	248	26609	81.39	78.29	80.51	88.75	87.02	87.33	91.28	92.72	**95.42**
Allotment	52	523	41.68	58.70	33.35	63.67	45.70	52.35	66.92	78.26	**78.39**
Commercial Area	7	1638	14.10	9.46	12.39	15.81	11.78	1.05	11.05	**22.55**	8.60
Water	23	1507	35.83	9.36	19.04	17.25	26.28	33.92	49.24	43.40	**52.16**
OA (%)			81.33	80.67	81.04	86.39	86.05	87.66	89.30	89.68	**92.05**
AA (%)			59.40	51.85	56.58	58.17	55.52	59.46	67.20	68.57	**71.79**
Kappa (%)			73.27	72.01	72.96	80.25	79.59	82.07	84.62	85.22	**88.59**

TABLE IV

CLASS-SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS ON THE CROSS-SCENE HOUSTON DATASET

Class Name	Training Samples	Test Samples	Classic Backbone Networks			Hybrid Backbone Networks					
			2-D CNN [54]	3-D CNN [21]	ViT [55]	MorphConv [56]	SSFTT [25]	WFCG [57]	DSNet [31]	HybridFormer [29]	S^2VNet
Grass Healthy	345	1353	76.42	68.07	37.77	78.12	**88.25**	53.73	73.32	87.21	82.93
Grass Stressed	365	4888	73.12	75.27	61.91	56.51	40.88	74.51	72.12	74.06	**82.59**
Tree	365	2766	44.61	32.28	29.93	**62.51**	35.61	36.26	54.66	35.47	41.94
Water	285	22	**100.00**	**100.00**	86.36	68.18	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
Commercial	319	5347	52.12	71.65	64.35	**78.55**	55.96	56.99	43.31	48.12	63.16
Non-Commercial	408	32459	88.59	84.93	87.03	81.63	89.21	90.46	**92.14**	90.72	87.64
Road	443	6365	55.82	54.36	14.19	58.08	19.83	29.19	**67.10**	38.24	65.34
OA (%)			76.99	75.89	69.51	75.10	70.32	74.55	78.57	75.67	**79.56**
AA (%)			70.10	69.51	54.51	69.08	61.39	63.02	71.81	67.69	**74.80**
Kappa (%)			60.57	59.64	44.70	59.62	45.02	54.72	64.52	57.85	**65.79**

article. The Adam optimizer is utilized to update the parameters of S^2VNet with a minibatch size of 64 on all datasets. The learning rate is set to 1e-3 and decays by multiplying a factor of 0.9 after each 50 epochs. The hyperparameters of the proposed method are empirically tuned to $\lambda_1 = 0.01$, $\lambda_2 = 0.01$, and $\lambda_3 = 0.01$. The number of heads is set to 4, and the embedding dimension C is 32. The dimension of the latent vector N_L in the general nonlinear AE unmixing network is set to 4, and the number of training epochs is 500. Except for patch-free WFCG, all samples of other classification methods are involved in training as hyperspectral patches. The patch size of 7×7 is adopted on Indian Pines and Augsburg datasets, ant the patch size of 5×5 is used on Berlin and cross-scene Houston datasets. The input data is normalized and scaled to [0,1] to accelerate the convergence speed.

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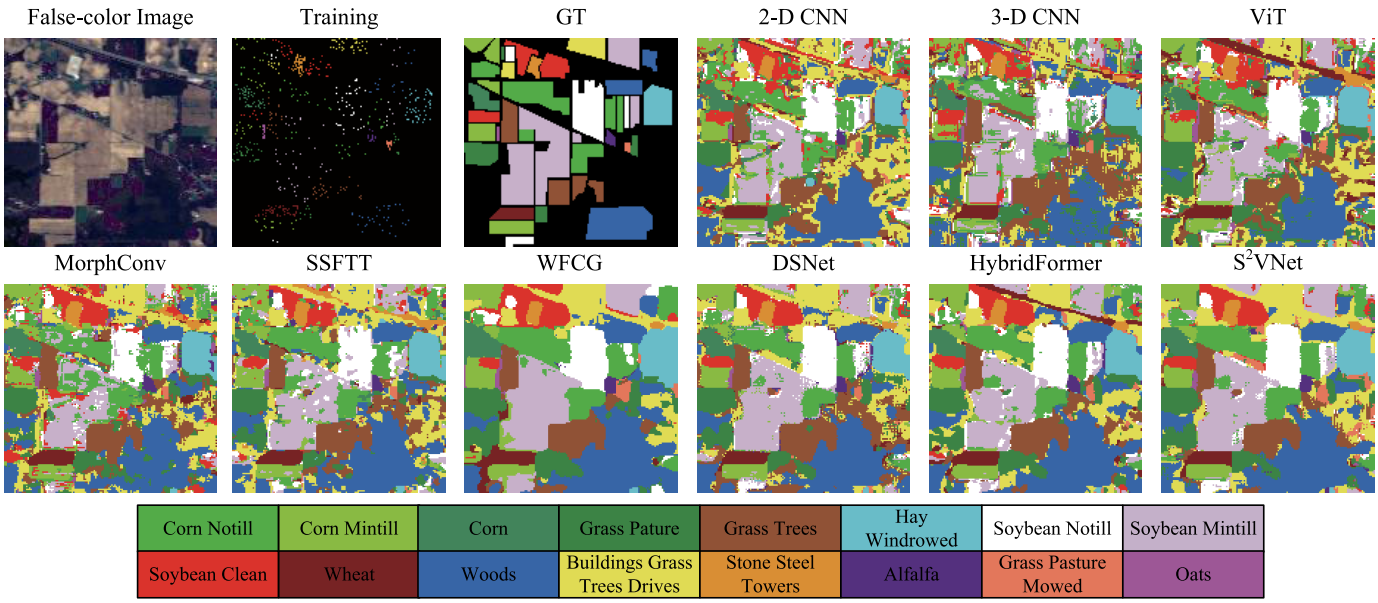


Fig. 2. False-color image, GT, and classification maps obtained by different methods on the Indian Pines dataset.

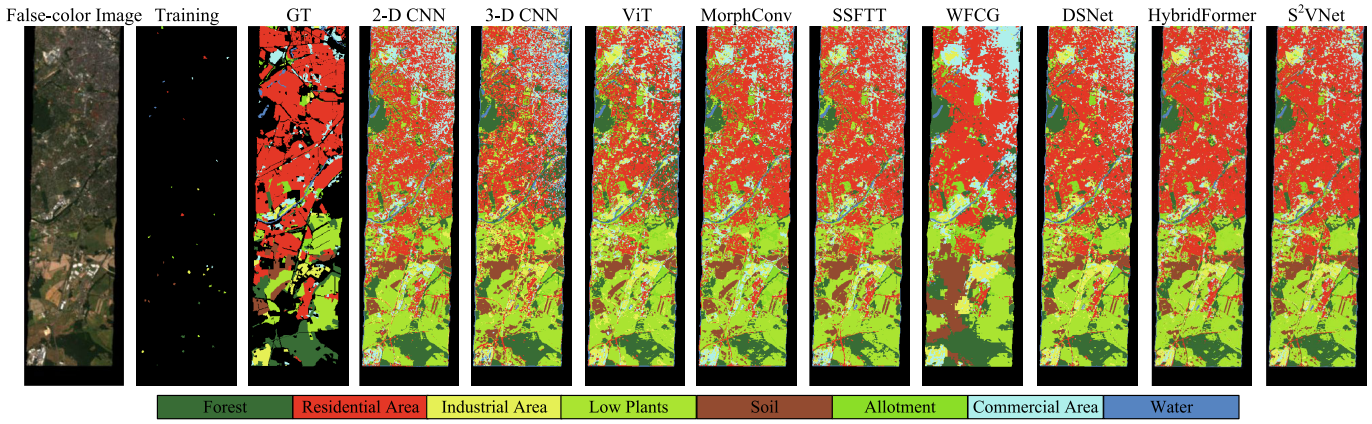


Fig. 3. False-color image, GT, and classification maps obtained by different methods on the Berlin dataset.

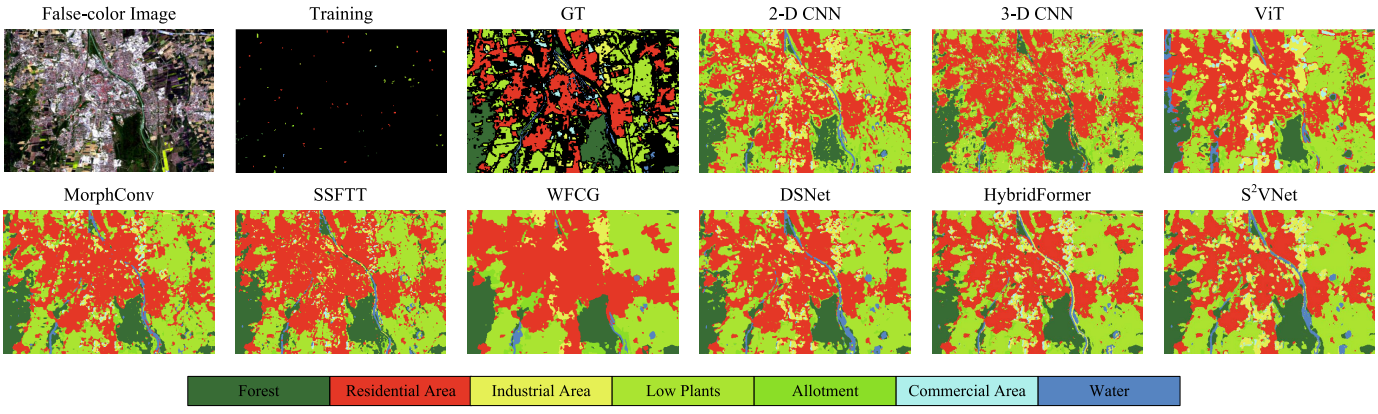


Fig. 4. False-color image, GT, and classification maps obtained by different methods on the Augsburg dataset.

B. Comparison With Current Methods

The quantitative and qualitative comparisons between our proposed S^{2}VNet and eight mainstream classification approaches are performed on four HSI datasets to assess the classification performance. Tables I–IV report the class-specific and overall classification accuracy results of different methods on all experimental datasets, including OA (%), AA (%), and Kappa (%). The highest accuracy for each evaluation metric is highlighted in bold. The corresponding false-color image, GT, and classification maps of various methods are displayed in Figs. 2–5. Overall, the proposed S^{2}VNet can obtain higher OA and Kappa results than other

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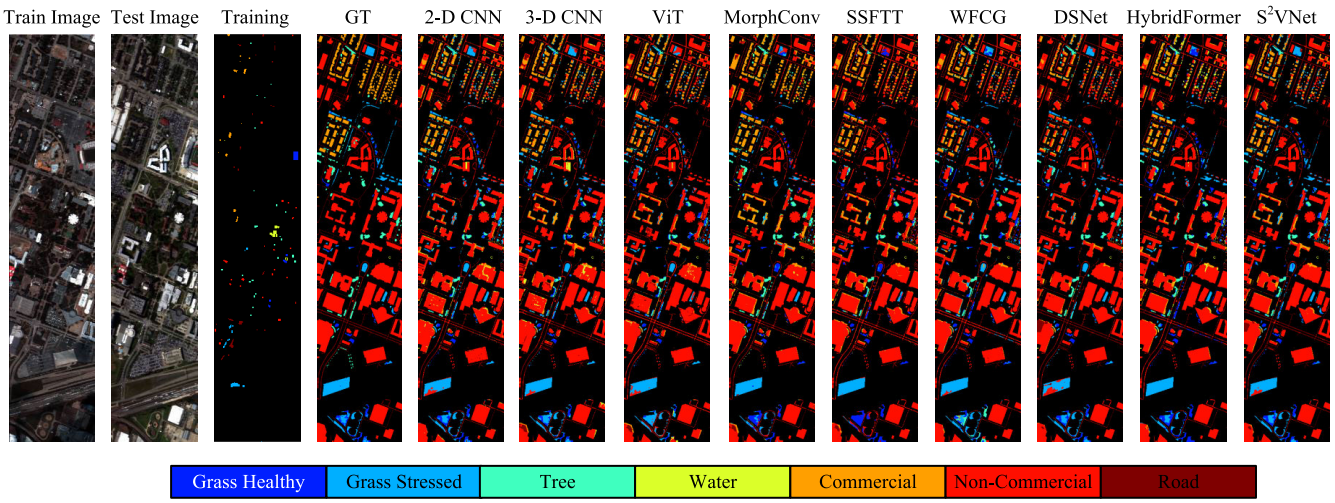


Fig. 5. False-color image, GT, and classification maps obtained by different methods on the cross-scene Houston dataset.

TABLE V
EXECUTION TIME (IN SECONDS) OF ONE EPOCH TRAINING IN DIFFERENT DL-BASED HSI CLASSIFICATION METHODS

Datasets	2-D CNN [54]	3-D CNN [21]	ViT [55]	MorphConv [56]	SSFTT [25]	WFCG [57]	DSNet [31]	HybridFormer [29]	S^{2}VNet
Indian Pines	0.57	4.99	2.14	4.39	0.96	**0.30**	0.75	4.57	0.83
Berlin	**28.93**	304.04	117.63	205.64	41.40	33.76	39.78	271.53	47.87
Augsburg	3.63	34.55	12.51	31.80	6.42	**3.24**	5.86	33.25	7.92
Cross-scene Houston	1.14	11.07	3.87	5.02	2.33	**0.91**	1.84	8.28	2.07

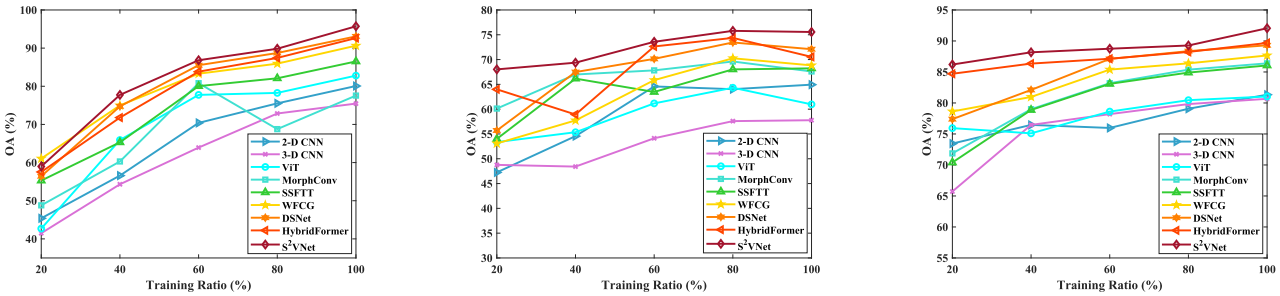
deep learning-based classification approaches on all datasets, which further demonstrates the effectiveness and superiority of the proposed method in real scenarios. In three classic backbone networks, the 2-D CNN can exhibit better classification accuracy and generalization performance compared with 3-D CNN and ViT, owing to the effective modeling of spatial and spectral information. Among hybrid backbone networks, DSNet and HybridFormer can obtain more competitive classification performance than MorphConv, SSFTT, and WFCG. The introduction of subpixel information and multigranularity spatial–spectral attention architecture enables the classifier to effectively exploit robust spatial and spectral information in the HSI. Compared with HybridFormer, it can be obviously seen from Table IV that certain self-supervised subpixel information can improve model generalization performance and maintain the optimal overall performance with 78.57% by extracting intrinsic high-dimensional features at the subpixel level. Different from DSNet, the proposed method considers the spectral cosine correlation information based on endmembers into subpixel fusion module and effectively reflects spectral changes for ground objects with similar spectral characteristics, such as three categories of Soybean Notill, Soybean Minta, and Soybean Clean on the Indian Pines dataset. As illustrated in Table I, the proposed method can significantly achieve better class accuracy than DSNet when dealing with similar spectral changes that existed in HSI scenarios. Furthermore, the proposed joint modeling of spectral variability and nonlinear mixture characteristics in S^{2}VNet can integrate more complete subpixel information and pixel-level class features to realize domain-invariant representation learning by approximately improving 2.64%, 3.47%, 2.75%, and 0.99% OA results over DSNet on different datasets, respectively. In addition, S^{2}VNet also achieves more accurate visualization results and preserves more details on various classes, as shown in Figs. 2–5. It should be noted that although the proposed method cannot obtain the ideal AA results over WFCG and HybridFormer on the Berlin dataset, it achieves optimal or suboptimal results for class 2 "Residential Area" and class 4 "Low Plants," which has a larger number of test samples on this dataset.

The one-epoch training time comparisons on four HSI datasets are listed in Table V to evaluate the computational complexity of various classification methods. It can be clearly observed that WFCG consumes the least time on Indian Pines, Augsburg, and cross-scene Houston datasets, while it cannot perform effectively on large datasets, for example, Berlin. WFCG considers the image-level superpixel and graph-based architecture to model HSIs and effectively reduce the computational complexity. The 2-D CNN method can achieve optimal computational efficiency in a large Berlin scene. Moreover, the architecture of self-attention and 3-D convolution enables HybridFormer and 3-D CNN to have the worst execution time on all datasets. Compared with the 2-D CNN, the introduction of AE unmixing architecture increases the computational complexity to a certain extent, but the overall difference is not large. Due to the complete spectral modeling and multilevel subpixel fusion design, the proposed S^{2}VNet takes more execution time than DSNet, but its performance is overall better than most hybrid backbone networks. In general, the computational time of the proposed method is acceptable on different HSI datasets while maintaining the best classification performance.

C. Performance on Different Ratios of Training Samples

To verify the generalization performance of the proposed S^{2}VNet and the other competitors, different ratios of training samples are selected from the given training set, and the

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(a) (b) (c)

Fig. 6. Classification performance of each method with different ratios of training samples on three datasets. (a) Indian Pines. (b) Berlin. (c) Augsburg.

corresponding classification performance on three datasets is displayed in Fig. 6. The OA indicator is used as the performance evaluation criterion, and the adopted training ratio is set as [20%, 40%, 60%, 80%, 100%]. In general, the OA results of all classification methods gradually improve the classification performance as the training ratio increases. When the training ratio is relatively small, at 20% on the Indian Pines dataset, the proposed S^{2}VNet method fails to achieve the optimal classification performance compared with WFCG. The main reason is that multiple ground object categories with fewer training samples cannot support the complete learning of spectral modeling, so the classifier is more inclined to pixel-level discrimination features. For scenarios with a small number of categories, such as Berlin and Augsburg, the proposed method exhibits superior performance among all classification algorithms. Compared with DSNet, S^{2}VNet considers more spectral characteristics and sufficient subpixel fusion features, which makes the classification results more robust under the circumstance of fewer samples. As the training ratio increases and exceeds 20%, the potential of the subpixel-level features can be fully exploited and the OA results of S^{2}VNet significantly outperforms other state-of-the-art classification approaches on all datasets, which further demonstrates its competitive classification performance and excellent stability.

D. Model Analysis

1) Ablation Study: To analyze the contribution of main components in S^{2}VNet, the ablation study with different combinations of network modules is investigated on three datasets, as listed in Table VI. The two primary components of S^{2}VNet are considered in the ablation study: enhanced subpixel fusion and unmixing network with $\hat{\mathbf{M}}_{sp}$. Note that the absence of enhanced subpixel fusion indicates that only the collaboration of the extracted abundance and pixel-level class feature is used to obtain classification results without attaching spectral cosine correlation information. The adopted unmixing network with $\hat{\mathbf{M}}_{sp}$ represents the joint modeling of spectral variability and nonlinear mixture characteristics to reconstruct the HSI in the proposed method. Different $\hat{\mathbf{M}}_{sp}$ settings indicate whether the spectral variability modeling introduces the nonlinear unmixing architecture. In the absence of an enhanced subpixel fusion module, only

TABLE VI

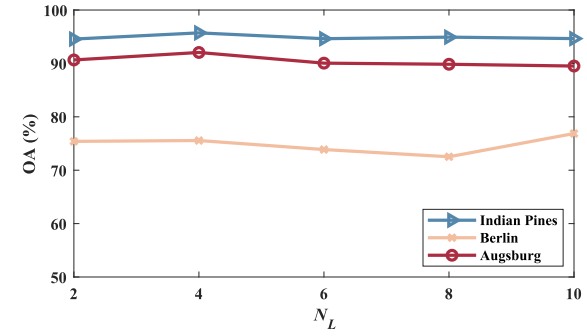
ABLATION COMPARISON OF EACH VARIANT IN S^{2}VNET

Module		Metrics	Datasets		
Enhanced Subpixel Fusion	With $\hat{\mathbf{M}}_{sp}$		Indian Pines	Berlin	Augsburg
✗	✗	OA (%)	90.72	70.97	86.75
		AA (%)	89.20	61.67	62.85
		Kappa (%)	89.38	57.56	81.11
✗	✓	OA (%)	93.60	74.15	89.95
		AA (%)	90.87	63.33	63.54
		Kappa (%)	92.67	60.68	85.39
✓	✗	OA (%)	93.70	73.07	90.14
		AA (%)	96.41	63.10	66.08
		Kappa (%)	92.78	60.14	85.83
✓	✓	OA (%)	**95.72**	**75.56**	**92.05**
		AA (%)	**97.77**	**63.88**	**71.79**
		Kappa (%)	**95.10**	**62.95**	**88.59**

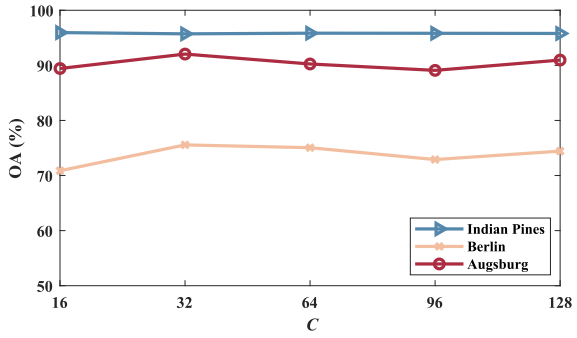
considering spectral variability modeling can increase the OA of the Indian Pines, Berlin, and Augsburg datasets by 2.88%, 3.18%, and 3.20%, respectively. Compared with considering the nonlinearity alone, the introduction of spectral variability modeling can achieve more complete subpixel information learning and enhance classification performance on different HSI scenarios. Although it is difficult to distinguish the degree of nonlinearity and spectral variability in practical application scenarios, the proposed method considers these two factors and adopts a data-driven architecture to effectively learn the complex changes of different ground objects. In addition, the introduction of an enhanced subpixel fusion module alone can exceed the baseline, respectively, by 2.98%, 2.10%, and 3.39% on each dataset, which can demonstrate the effectiveness of the joint exploitation of subpixel-level and pixel-level features in S^{2}VNet. Moreover, the combination of these two components can improve classification performance more precisely than one component alone, further illustrating the value of complete subpixel modeling and feature fusion for mining significant information in actual HSI scenarios.

2) Parameter Tuning: The effect of two tuning parameters in the proposed method is analyzed, including the dimension of latent vector N_L and the embedding dimension C. N_L controls the learned pattern of spectral variability in the latent space. A large value of N_L can provide richer variability modeling but have high model complexity in the optimization process. C represents the low-dimension feature space for the encoded patch token in the pixel-level classifier network. Fig. 7 illustrates the obtained OA results with different N_L and C values on three HSI classification datasets. The adjustable ranges of N_L and C are set as

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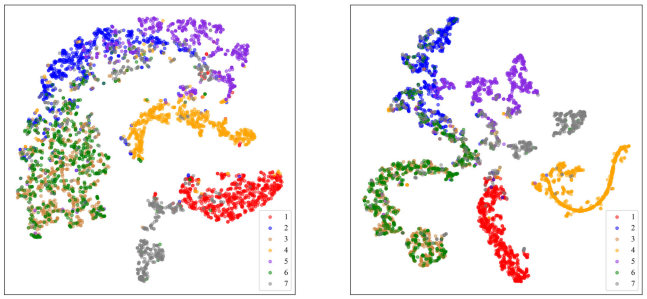


(a)



(b)

Fig. 7. Classification performance analysis between OA and various parameters on different datasets. (a) Dimension of latent vector N_L. (b) Embedding dimension of pixel-level classifier C.



(a) (b)

Fig. 8. t-SNE embedding visualization of class features obtained by DSNet and S^{2}VNet on the Augsburg dataset. (a) Class features output by DSNet. (b) Class features output by S^{2}VNet.

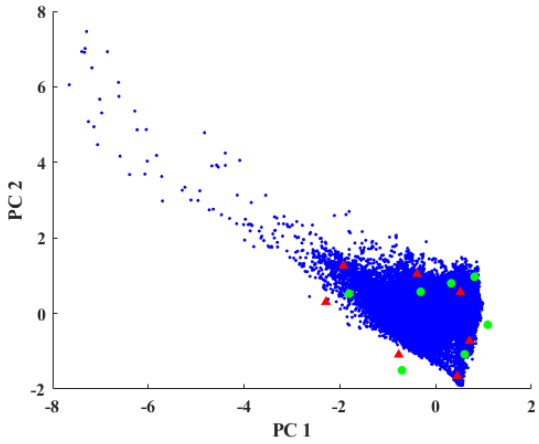


Fig. 9. Illustration of scatter plot after projection onto two principal components (PCs).

{2, 4, 6, 8, 10} and {16, 32, 64, 96, 128}, respectively. It can be seen from Fig. 7 that the proposed method with parameter N_L equal to 2 and 4 can achieve suboptimal classification performance on different HSI datasets. When N_L is set to 4, the proposed method has the best classification performance, and the spectral variability modeling in this latent dimension is suitable for providing robust subpixel feature estimation in the general nonlinear AE unmixing network. For the embedding dimension C, it presents parameter insensitivity on the small-scene Indian Pines dataset and the corresponding classification performance tends to be stable. The main reason for the small changes is that the spectra of different ground objects are relatively distinguishable on the Indian Pines dataset, and each category can be well learned in S^{2}VNet. On Berlin and Augsburg datasets with more labeled samples, the proposed method exhibits a certain degradation of classification performance as the value of C increases because of the introduction of excessive encoding feature dimensions. When the embedding dimension is 32, the encoded patch token is more adequate for the pixel-level classifier training and subsequent fusion process. Therefore, in all experiments, the values of N_L and C are uniformly fixed to 4 and 32 for a fair comparison.

3) Visual Evaluation: For illustrative purposes, the t-distributed stochastic neighbor embedding (t-SNE) [61] is utilized to visualize the class separation performance of different classification methods. Fig. 8 displays the t-SNE embedding visualization of class features obtained by DSNet and S^{2}VNet on the Augsburg dataset. Both methods consider subpixel information extracted from the HSI but focus on different spectral modeling problems in the HSI classification task. It can be clearly seen that the interclass distribution of DSNet is still mixed up, for example, 2th, 4th, and 5th classes. The main reason is that DSNet tends to generate accurate abundance information as subpixel prior and ignore spectral heterogeneity existing in the HSI. On the contrary, the introduction of spectral variability modeling helps S^{2}VNet improve the separation degree of different categories, such as the 2nd and 5th classes as well as the 4th and 5th classes. For relatively similar land-use categories, such as the 2nd, 3rd, and 6th classes, DSNet cannot achieve a good separation for the three categories. Although few labeled samples limit the separation performance between the 3rd and 6th classes, S^{2}VNet is more inclined to learn spectral differences of the 2nd class than the other two categories. Moreover, the designed fusion module enables the proposed method to consider the complete multilevel class information and better cluster the same category. Therefore, the distribution of class features obtained by S^{2}VNet presents a more reliable decision boundary compared with DSNet.

To validate the effectiveness of spectral variability modeling, the illustration comparison of the estimated endmembers for these two methods are displayed via projection onto a 2-D space in Fig. 9. The blue dots represent the data samples after projection. The green circles and red triangles correspond

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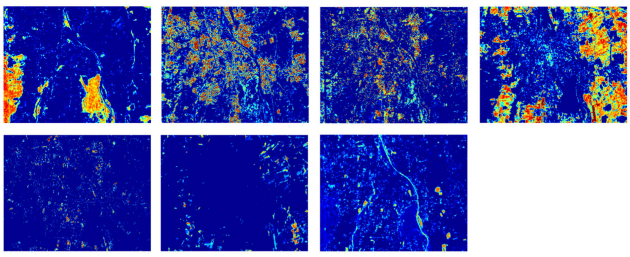


Fig. 10. Visualization of the extracted abundance maps obtained by S^{2}VNet.

to the estimated endmember results of DSNet and S^{2}VNet, respectively. It is apparent that the endmember estimates of the proposed method can cover more distant data samples and capture rich variation patterns because of flexible variability modeling of scaling factors and perturbation terms. However, DSNet lacks consideration of spectral variability modeling, which further restricts the distance between different endmembers. In addition, the extracted abundance maps obtained by S^{2}VNet is shown in Fig. 10. Due to the lack of abundance GT on the Augsburg dataset, we only conduct a qualitative analysis of abundance maps for different ground objects. It can be seen that the extracted abundance maps can significantly indicate the distribution information of various categories, such as class 2 "residential area" and class 3 "industrial area." Overall, the proposed method can obtain accurate endmember and abundance results with the help of complete spectral modeling and multilevel fusion strategy, which further demonstrates its effectiveness in practical scenarios.

V. Conclusion

This article proposes a S^{2}VNet framework for HSI classification tasks, which effectively deals with the mixed pixel and spectral heterogeneity problem in actual scenarios. By leveraging general nonlinear AE unmixing architecture to simultaneously model variability parameters and nonlinear mixture characteristics, S^{2}VNet can obtain complete subpixel information to ensure accurate subpixel-level feature estimation. The enhanced subpixel fusion module is further designed to aggregate three parts of abundances, spectral cosine correlation information, and pixel-level class features to provide more complete joint representations for the classifier training. Extensive experiments on four benchmark datasets validate the effectiveness and superior generalization of the proposed S^{2}VNet compared with several state-of-the-art classification methods. In future work, we would like to explore more lightweight network architecture designs and advanced applications for orbit real-time HSI processing.

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